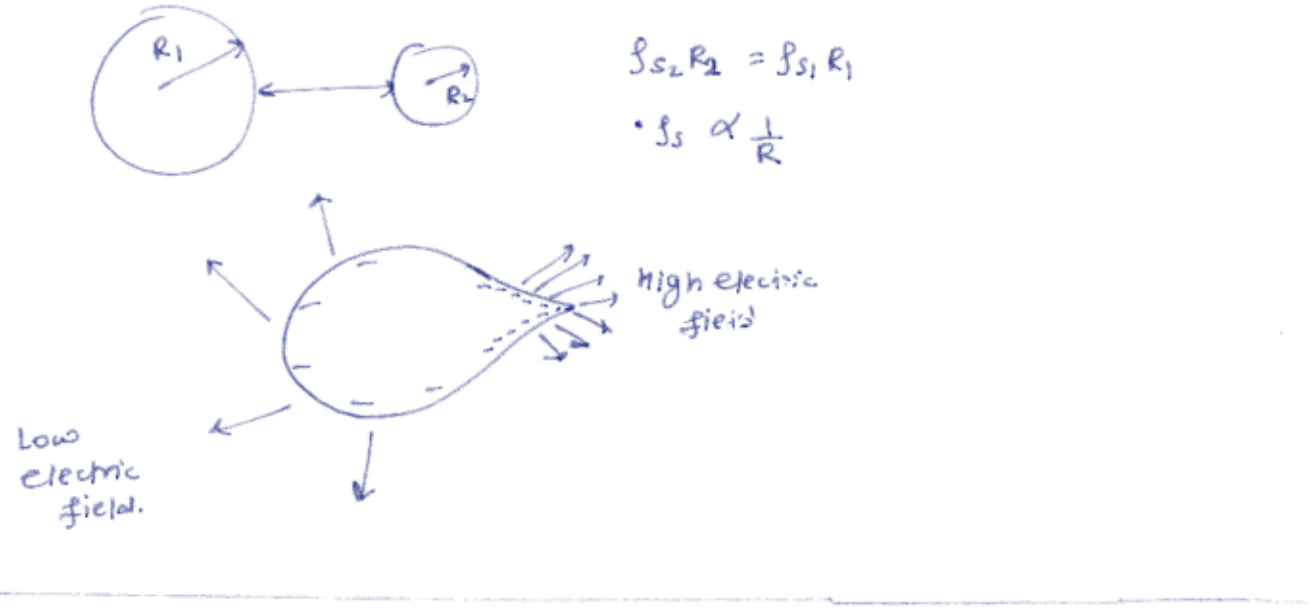
Corona

CORONA :

(1) Corona is a self electrical discharge of atmospheric air.

(2) If the diameter of the conductor is of small value, this will increase the charge density so that electric field is increased.

(3) If this electric field is more than dielectric strength of atmospheric air, then air molecules are ionized, which initiates corona.



Two spheres R1 and R2: charge density x radius constant, rho_s inversely proportional to R; charge crowding at the pointed end gives high electric field, low field elsewhere

$$\rho_{s2} R_2 = \rho_{s1} R_1 ; \quad \rho_s \propto \frac{1}{R}$$

Critical Disruptive Vtg (V_c)

The Vtg at which corona is just initiated is called V_c. At this Vtg corona is not visible but sound can be heard.

$$V_c = m_o g r \delta \ln\left(\frac{d}{r}\right) \quad kV/ph \text{ (rms)}$$

m_o = Surface irregularity factor = 0.8 to 0.9 for ACSR (Rough conductor) ; = 0.9 to 1 for Hollow / Smooth conductor

g = Dielectric strength of Atm. Air = 21.1 kV/cm (rms)

r = radius of conductor in (cm) ; d = distance between conductors in cm

δ = Air density factor = $\frac{3.92b}{273+T}$

T = Atm. Temp in $^\circ C$; b = Atmospheric air pressure in cm of Hg.

Critical visual Vtg (V_v) :

The Vtg at which corona is visible with violet or blue colour is called V_v.

$$V_v = m_v g r \delta \left(1 + \frac{0.3}{\sqrt{r\delta}}\right) \ln\left(\frac{d}{r}\right) \quad \text{kV/ph (rms)}$$

m_v = Surface / Irregularity factor corresponding to V_v.

Corona power loss (P_L) :

If $V_{ph} > V_c$ corona occurs and produces power loss.

$$P_L = 242.2 \times 10^{-5} \frac{(f+25)}{\delta} \sqrt{\frac{r}{d}} (V_{ph} - V_c)^2 \quad \text{kW/km/ph}$$

Before applying the P_L formula check the following conditions :

Step 1: Calculate V_c. If $V_c < V_{ph}$ then only calculate P_L.

Step 2: $V_c \geq V_{ph}$ then $P_L = 0$

$$V_c = m_o g r \delta \ln\left(\frac{d}{r}\right)$$

Factors affecting corona and methods of reducing corona :

* $V_c > V_{ph}$ corona doesn't occur, so that $P_l = 0$ and hence in practical applications higher V_c is maintained by changing the design parameters. $V_c \propto m_o r$

* m_o and r values are higher in case of hollow conductor and hence V_c is higher, corona is less compared to the ACSR.

* Isolators are made of hollow conductor in the S/S to minimize corona.

* Bundled conductors have higher self GMD so that V_c is higher, corona is less.

* $V_c \propto \delta$, $V_c \propto b$: V_c is less in case of hilly area compared to plain area and hence corona is more in hilly areas.

* $V_c \propto \ln(d)$: By maintaining optimum phase distance between P-P and P-E wire, higher V_c is maintained to minimize corona.

* By placing corona ring at the edges of T/L conductors, surface area is increased so that charge density and electric field are reduced and hence corona is minimized.

* $P_l \propto f + 25$: HVAC has higher corona because freq. is higher compared to HVDC.

Advantages :

Steepness of the surge due to lightning is minimized.

Disadvantages :

* Power loss increased * Charging current is increased

* Radio interference is increased due to harmonics produced during corona.